carries in a pouch on the front of its body. Kangaroos are herbivores and have chambered stomachs similar to those of cattle and sheep. They regurgitate the vegetation they eat, chew it as cud and then swallow it again for final digestion.

KANTOROVITCH INEQUALITY: The inequality, valid for any positive definite matrix, P and any non-zero vector, x, that $\frac{\langle x,x\rangle\langle x,x\rangle}{\langle x,\mathrm{P}x\rangle\langle x,\mathrm{P}^{-1}x\rangle}=\frac{4mM}{\langle m+M\rangle^2}$, where m and M are respectively, the smallest and the largest eigenvalues of P. This is useful in estimating convergence rates for descent methods.

KAOLIN (CHINA CLAY): A fine, soft white clay, composed mainly of the mineral kaolinite, formed during the weathering and hydrothermal alteration of other clays or feldspar. In its natural form, it is a white, soft powder consisting principally of the mineral kaolinite. Removal of small amount of silicon dioxide or silica (SiO_2) and mica enables it to be mixed with water to give white plastic clay. It is used in making porcelain, medicines, ceramics, paper, rubber, paint and many other products. **Kaolin group** is hydrous aluminosilicates having a 1:1 phyllosilicate, structure and no permanent charge.

KAOLINITE: A white or grey clay mineral which is the chief constituent of kaolin, with the chemical formula $Al_2Si_2O_5(OH)_4$. It is the mineral form of aluminium potassium silicate ($KAlSi_3O_8$); found in sediments, soils, hydrothermal deposits and sedimentary rocks and formed by the alteration of other minerals, especially feldspar, which is the most common constituent of kaolin. It is a member of a group of clay minerals called the kaolin group minerals. It is an important industrial commodity used in ceramics, paper coating and filler, paint, plastics, fibreglass and catalysts.

KAPOK (KAPOK TREE; SILK COTTON TREE; *Ceiba pentandra*): An emergent layer tree of the tropical forest belonging to the Malvaceae family of the genus Ceiba, found in Africa and parts of Asia and Indonesia where the floss is an important product of Java. It grows up to about 30 - 40 metres in height, although some varieties can grow up to 70 metres. It has huge buttressed trunk that tapers upwards to an almost horizontal, spreading crowns with compound leaves made up of five to eight long, narrow leaflets. As a deciduous plant, it sheds its leaves during the dry season paving way for flowering. This improves access for bats to feed on the sugar-laden nectar and also pollinate the flowers. When the tree blooms it can produce up to 4,000 fruits. Kapok timber is used for a variety of wood products, including paper. Some locals use kapok logs for carving canoes.

KAPOSI'S SARCOMA (KS): Malignant tumour developing from blood vessels in the skin that occurs mainly in people with depressed immune systems, for example, as a result of AIDS. It mainly affects the skin, mouth and lymph nodes, but can also affect other parts of the body, such as the lungs or digestive tract. It is caused by infection with human herpesvirus 8 (HHV8).

KARMARKAR METHOD: A polynomial time algorithm for linear programming that is based on projective transformations and is of the nature of interior penalty function methods. This appears to compete well with algorithms based on simplex method for certain classes of programs.

KARNAUGH MAP (K-MAP): Introduced in 1953 by Maurice Karnaugh, an IBM Engineer, it is a modified truth table used for the minimisation of Logic functions. In a Karnaugh Map, the output column is redrawn as a two dimensional array of cells or boxes, each containing a '0' or '1' entry. The 1-cells correspond to minterms and the 0-cells to maxterms.

KARST: A ground typical of limestone with an uneven surface, holes and cracks due to weathering. It is characterised by fissures called grikes or grykes, depressions called dolines or dolinas and underground streams.

KARUSH-KUHN-TUCKER THEOREM (KUHN-TUCKER CONDITIONS): A result extending the Lagrange method of multipliers to constraints by equalities and inequalities. It ensures that, given an appropriate constraint qualification, any point x_0, that minimises $f(x)$ subject to $g_1(x)\le 0,\ldots,\ g_n(x)\le 0, g_{1+n}(x)=0,\ldots,g_{m+n}=0$ will occur at a stationary point of the Lagrangian $\mathrm{L}(x, \lambda) = f(x) + \Sigma\lambda_i g_i$, where the summation is over the binding constraints with $g_k(x_0)=0$ and where the multipliers corresponding to inequality constraints are nonnegative.

KARYOGAMY: The fusion of cell nuclei during fertilisation or asexual reproduction. It is the major mode of reproduction in fungi.

KARYOGRAM: A graphical representation or diagram of chromosome, generally arranged in pairs and in order of size. It shows the characteristic features of the chromosomes of a species.

KARYOKINESIS: The division of cell nucleus, which occurs during meiosis or mitosis.

KARYORRHEXIS: A stage of cell death that involves destructive fragmentation of the nucleus. The nucleus breaks down into small dark beads of damaged chromatin. This usually leads to further dissolution of the nucleus and the gradual fading of the damaged chromatin.

KARYOTYPE: The number and structure of the chromosomes in the nucleus of a cell, constituting the entire genome o of an organism. The karyotype is identical in all the diploid cells of an organism. The human karyotype contains forty-six chromosomes, twenty-three from each parent; in mice 40, crayfish 200; and in fruit flies 8.

KASTLE MEYER TEST (PHENOLPHTHALEIN TEST): A forensic presumptive test to indicate the presence of blood. Phenolphthalein and hydrogen peroxide are used to react with haemoglobin in the blood to give it a pink colour.

KATABATIC WIND: A cool wind that blows on calm clear nights carrying high-density air from a higher elevation down a slope under the force of gravity. By contrast, an anabatic wind is warm and moves up a valley in the early morning.

KATADROMOUS (CATADROMOUS): The migration of certain fish such as the freshwater eel (*Anguila anguila*) that spend most of their lives in freshwater before swimming into deep oceanic water to breed.

KATAL: A derived SI unit of enzyme activity for expressing quantity values of catalytic activity of enzymes and other catalysts. It is defined as the catalytic activity of an enzyme that increases the rate of conversion of specified chemical reaction by 1 mol under specified assay conditions. Its symbol is kat. For example, one katal of trypsin, is the amount of trypsin which breaks a mole of peptide bonds per second under specified conditions.

KATER'S PENDULUM: A reversible free swinging pendulum invented by British physicist, Henry Kater (1777-1835) in 1817 for measuring the acceleration of free fall or gravity. It is made of a metal bar with knife edges attached near the ends and two weights that can slide between the knife edges. The bar is pivoted from each knife edge in turn and the positions of the weights are adjusted so that the period of the pendulum is the same with both pivots. The period is then given by the formula for a simple pendulum, which enables gravity to be calculated.

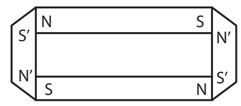
KATETOV'S INTERPOLATION THEOREM: The result that if a lower semi-continuous real-valued function, f, majorises an upper semi-continuous real valued function, g, both with domain a normal topological space, then there is a continuous function, h, such that $f(x) \ge h(x) \ge g(x)$.

KATHAROMETER: An instrument for gas analysis using thermal conductivity measurements by comparing the rate of loss of heat from two heating coils surrounded by the gases. The thermal conductivity of a gas is inversely related to its molecular weight. It can be used to detect the presence of a small amount of an impurity in air.

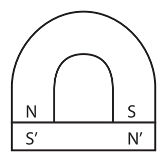
KAWASAKI DISEASE (KAWASAKI SYNDROME; MUCOCUTANEOUS LYMPH NODE SYNDROME): A rare childhood acute inflammatory disease that is one of the leading causes of acquired heart disease in children in which the walls of the blood vessels throughout the body become inflamed. The disease can affect any type of blood vessel in the body, including the arteries, veins and capillaries. Its symptoms include prolonged fever, congestion of the conjunctiva of the eyes, changes in the lips and oral cavity, swelling of the cervical lymph nodes, skin rash, reddening of the palms of the hands and soles of the feet and damage to the coronary arteries. The cause of Kawasaki syndrome remains unknown.

KED (SHEEP TICK; SHEEP KED): A wingless parasitic fly, *Melophagus ovinus*, of sheep that causes extreme skin irritation leading to loss of wool.

KEEL: 1. Also known as **carina**, it is a bone projecting from the sternum (breast bone) of a bird or bat, to which the major powerful muscles are attached for flight. 2. The two lower petals of a papillonaceous flower that are fused together. A papilonaceous flower belongs to the pea family. 3. A prominent longitudinal ridge, resembling the lengthwise timber or steel structure along the base of a ship. Keeled means having ridges.



Keepers for bar magnets

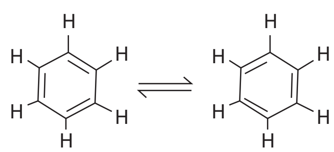


Keeper for horse-shoe magnet

Keeper

KEEPER: Soft iron magnets, which are used to cover the ends of a permanent magnet when they are not in use. The region near the ends of the magnet produce an induced flux within it opposing the original magnetising flux. The flux induced in the keepers is opposite in direction is that in the magnet itself and thus the demagnetising effect in the magnet is neutralised.

KEKULÉ STRUCTURE (KERKULÉ FORMULA): A representation of a benzene molecule as a hexagonal ring with alternating single and double bonds linking a six-member ring of carbon atoms, each linked to one hydrogen atom at the vertices. It was proposed by German physicist, Friedrich August Kekulé (1829-1890). It accounted for the molecular formula C_6H_6 and the reason for benzene yielding only one mono-substituted derivative and three de-substituted derivatives. However, it could not account for the unusual stability of benzene, why all the bond lengths are equal and why benzene resists addition reaction and common oxidising agents.



Kekulé structure of benzene

KELP: Any large leafy brown sea weed or algae, Phaeophyta belonging to the order Laminariales that grows along colder coastlines. They are used for extraction of alginic acid and also used as food or green manure and as a source of sodium.

KELVIN: Another name for the absolute temperature scale, with the symbol K. It was named after the British physicist and engineer Lord William Thompson Kelvin (1824-1907). The temperature of a substance at absolute zero is zero Kelvin (0 K). The Kelvin and the degree Celsius are often used together, as they have the same interval and 0 K = -273.15 °C; 273.15 K = 100 °C.

KELVIN BALANCE (AMPERE BALANCE; CURRENT BALANCE): An electromechanical device used to measure current absolutely, on the basis of definition of the ampere.

KELVIN SCALE: The basic scale used universally by scientists for temperature definition. The triple point of water comprising ice, liquid and vapour is defined as 273.15 K. The Kelvin scale begins at absolute zero (-273.15 °C) and increases at the same degree intervals as the Celsius. The Kelvin and the degree Celsius are often used together; 0 K = -273.15 °C and 373.15 K = 100 °C.

KELVIN, LORD WILLIAM THOMPSON (1824-1907): British physicist, who was born in Belfast. He entered the University of Glasgow at age 10, published two papers by 17, graduated from Cambridge University at 21 years old and became professor of natural philosophy at Glasgow University in 1846. He carried out important experimental work on electromagnetism, inventing the mirror galvanometer and contributing to the development of telegraphy. He worked with James Joule on the Joule-Thomson effect. His main theoretical work was in thermodynamics, in which he stressed the importance of the conservation of energy. In 1848, he invented the absolute temperature scale named after him. The unit of thermodynamic temperature is also named after him. He also contributed to the determination of the age of the Earth and the study of hydrodynamics. In 1896, he was created Baron Kelvin of Largs in South Western Scotland.

KELVIN'S CIRCULATION THEOREM: The theorem that the circulation of a frictionless fluid is invariant in time in the presence of conservative forces. The theorem also holds for a fluid of uniform density that is not compressible. It was published in 1869 by William Thomson, 1st Baron Kelvin and named after him. It is defined mathematically as $\frac{D\Gamma}{Dt}=0$, where Γ is the circulation around a material contour C(t).

KENAF (BROWN INDIAN HEMP): A fibre producing plant, *Hibiscus cannabinus*, in the Malvaceae family similar to jute. Its uses include in making various paper and pulp products, coarse fabrics, rope, as absorbents for chemical and oil spills and packing materials.

KENDELL METHOD: A method in statistics that is used in measuring the degree of association between two different ways of ranking n objects, using two variables (x and y) which give data $(x_1, y_1), \ldots, (x_n, y_n)$. The objects are ranked using the first x and the y. For each of the $\frac{2n}{\frac{1}{2}(n-1)}$ pairs of objects a score is assigned. Kendell's coefficient of rank correlation is given by $\tau = \frac{sum\ of\ sources}{\frac{n}{2}(n-1)}$. The closer τ is to one, the greater the degree of association between the ranks.

KENNEDY SPACE CENTRE (KSC): US National Aeronautics and Space Administration (NASA) launch site on Merritt Island, near Cape Canaveral, Florida, used for Apollo project and space shuttle launches. The first Apollo flight to land on the Moon (1969) and Skylab, the first orbiting laboratory (1973), were launched from the site. It was named in honour of President John F. Kennedy.

KEPLER, JOHANNES (1571-1630): German astronomer, who was educated at the University of Tübingen. In 1594, he became a mathematics teacher in Graz, where he learned of the work of Copernicus. He developed a theory that the cosmos was constructed of the five regular polyhedrons, enclosed in a sphere, with a planet between each pair. In 1600, he went to work for Tycho Brahe in Prague after sending his paper on the subject to him. It was there that he worked out Kepler's laws of planetary motion, for which he is best remembered. He was the first person to explain accurately, how light behaves within the eye, how eyeglasses improve vision and what happens to light in a telescope. In 1609, he published his finding that the orbit of Mars was an ellipse and not the perfect circle as used to be believed to be the orbit of every celestial body. This fact became the basis of the first of Kepler's three laws of planetary motion. The second law states that planets move faster as they near the Sun. In 1619, he showed that a simple mathematical formula related the planets' orbital periods to their distance from the Sun, his third law.

KEPLER'S LAWS OF PLANETARY MOTION: The three laws devised by Johannes Kepler to define the mechanics of planetary motion stating that: (i) each planet travels in an ellipse with the Sun being one focus of the ellipse; (ii) the radius vector from the sun to the planet sweeps out equal areas in equal times. This makes the planet moves quickest when the vector radius is closest to the Sun and moves more slowly when the radius vector is furthest from the Sun; (iii) the squares of the periodic times of the planets are proportional to the cubes of the semi-axes of the elliptic orbit. It was named after German mathematician, astronomer and astrologer Johannes Kepler (1571-1630).

KERATIN: A strong fibrous protein forming the outer protective layer of skin of vertebrates and also in hair, nails, claws, hoofs, feathers and the outer coatings of horns in animals such as cows and sheep. Keratin does not dissolve in cold or hot water and does not easily undergo proteolysis. **Keratinisation** is the process by which the outermost cell of the mammalian epidermis is replaced by keratin or the conversion of skin cells into other horny materials such as scales or nails.

KERATITIS: An eye disease characterised by inflammation of the cornea. This may be caused by an infection of bacteria, viruses, fungi or parasites. It may also be caused as a result of minor injury, for example, being hit in the eye, contact with chemicals or putting on contact lenses for a long time. Its signs and symptoms include red and painful eye, excess tears or other discharge from the eye, blurred and decreased vision, sensitivity to light and difficulty in opening the eye. If it persists for a long time without treatment, it can lead to serious complications, such as vision loss.

KERMIT: 1. A simple File Transfer Protocol that is used extensively for Pc-to Pc communications. 2. A type of data link protocol whose function depends on the Physical Separation (d) between two communicating Data Terminal Equipment and the bit rate (R) of the link. 3. An example of a character-oriented idle RQ (Stop and Wait) Protocol used for lower bit rate links such as those used with modems.

KERNEL: 1. The inner softer edible part of a nut or a seed within a hard shell. It forms the reproductive structure in plants. It consists of a plant embryo, surrounded by endosperm produced during fertilisation and enclosed in a protective coat. The endosperm provides nourishment for the plant embryo. 2. A positively charged atomic nucleus that has lost its valence electrons. 3. The set of elements of the domain of a mapping that have the identity element of the range as their image. The kernel of a homomorphism from a group to another is a normal subgroup; the kernel of a ring homomorphism is the counterimage of zero and is an ideal. 4. A function whose product with a given function is integrated to produce an integral transform of the given function. In the equation $g(t)=\int_a^b K(s,t)f(s)\mathrm{d}t$ the kernel is K. 5. **Iterated kernels** are the sequence of kernels defined by $K_0 = K$ and $K_{n+1}(s,t)=\int_a^b K(s,r)K_n(r,t)\mathrm{d}r$. These arise in the solution of the integral equations, with the sum $-\sum_{n=0}^{\infty}\lambda^n k_{n+1}(s,t)$ being called **resolvent kernel**. 6. The core of a computer's operating system that includes the most heavily used portion of software. The kernel is maintained permanently in main memory and performs essential system functions such as controlling memory and files and allocating system resources.

KERNING: A term used in typography to describe the spacing between the characters of a font. It adjusts the space between pairs of letters to make them more visually appealing. When kerning is applied to a font, the characters can vertically overlap and allows part of two characters to take up the same vertical space. This allows more text to be placed within a given amount of space. Kerning is not usually done with body text because the gaps between characters at body text sizes are generally not as distracting.

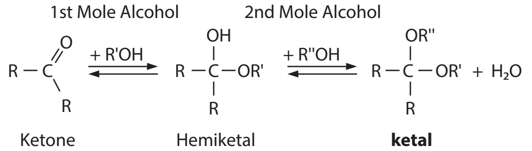
KEROGEN: Aged and degraded carbon that was once assimilated by living organisms and is non-extractable using organic solvents. Kerogen consists mainly of paraffin hydrocarbons, though the solid mixture also incorporates nitrogen and sulfur. Kerogen is insoluble in water and in organic solvents such as benzene or alcohol. Upon heating under pressure, however, the large paraffin molecules break down into recoverable gaseous and liquid substances resembling petroleum. In contrast, the fraction that is soluble in organic solvents is called bitumen. Kerogen represents about 90% of the organic carbon in sediments. It can be classified by its source material as Type I, consisting of mainly algal and amorphous (but presumably algal) kerogen and highly likely to generate oil; Type II, mixed terrestrial and marine source material that can generate waxy oil; and Type III, woody terrestrial source material that typically generates gas.

KEROSENE (KEROSINE; PARAFFIN; PARAFFIN OIL; COAL OIL): Colourless flammable oil distilled from petroleum and also derived from coal tar or oil-bearing shale. It can be separated physically from the other crude oil fractions by distillation or it can be produced by chemically decomposing or cracking, heavier portions of the oil at elevated temperatures. It is slightly heavier than gasoline and reaches its boiling point between about 180° and 290 °C. It contains hydrocarbons with 10 to 16 carbon atoms per molecule. Kerosene is less volatile than petrol, making it a relatively safe fuel to store and handle. Its flash point is 38 °C or higher, whilst that of petrol is as low as -40 °C. It is used as a fuel for jet engines, for heating, cooking and lighting. Standard commercial jet fuel is essentially a high-quality straight-run kerosene and many military jet fuels are blends based on kerosene.

KERR CELL: A transparent glass cell containing a dielectric liquid that exhibits the Kerr effect when light passes through. It consists of a small sealed glass cell into which two metal electrodes are inserted. The cell is filled with pure nitrobenzene, a transparent liquid, which exhibits double refraction when a high potential difference is applied across the electrodes. It is often used in devices where the intensity of light is to be changed rapidly in accordance with the voltage applied to the electrodes. For example, it is used to modulate light in a high-speed camera shutter.

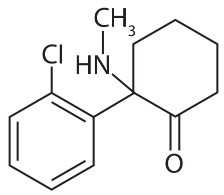
KERR EFFECT: 1. The ability of certain transparent substances to refract differently light waves whose vibrations are in two directions when the substance is placed in an electric field. 2. The elliptical polarisation of plane-polarised or unpolarised light when reflected from the polished pole of a magnetised material.

KETALS: A type of acetal compound with the general structure $R_2C(OR')_2$ where neither of the R group is a hydrogen atom. It is an organic compound formed when two moles of alcohol react with the carboxyl carbon of a ketone. In other words, it is a functional group containing two alkyl groups and two alkoxy groups on one carbon atom; produced in the acid-catalysed alcoholysis of a ketone or a hemiketal. An example of a ketal compound is cyclohexanone diethyl ketal. The compound formed when one mole of alcohol reacts with the carbonyl carbon of a ketone is called **hemiketal**.



Formation of a ketal

KETAMINE: A white crystalline powder with the chemical formula $C_{13}H_{16}ClNO$ and melting point 262 °C (535 K). It is used as anaesthetic in humans and veterinary medicine. It produces hallucinogenic effects and is used illegally in clubs.



Chemical structure of Ketamine

KETENES: A class of compounds of the type $R_1R_2C{=}C{=}O$, where R_1 and R_2 are organic groups. It is formed by dissolving a ketone in methanol and hydrochloric acid. An equilibrium is established which results in the formation of dimethylketal. Ketenes are reactive compounds and are often generated in a reaction medium for organic synthesis. The compound with a molecular structure $CH_2{=}C{=}O$ called **ethenone**. It is the simplest member of the ketene class. It is a tautomer of ethynol.

KETO ACID: A compound that is both an acid and a ketone. An example is acetoacetic acid ($CH_3CO.CH_2COOH$).

KETO-ENOL TAUTOMERISM: The existence of a chemical compound in two double-bonded structural forms, keto and enol, which are in equilibrium. The keto form (–$CH_2C{=}O$) changes to enol (-CH=C(OH-) by the migration of a hydrogen atom to form a hydroxyl group with the oxygen of the carbonyl group. The position of the double bond also changes.

KETOALDONIC ACIDS: Monosaccharides in which a structure containing a keto group and a carboxylic acid group is in equilibrium with a hemiacetal structure. Specific compounds are named using the -ulosonic acid suffix, for example, D-*arabino*-hexulosonic acid.

KETOALDOSES: Monosaccharides which contain both an aldehydic and a ketonic carbonyl group in equilibrium with intramolecular hemiacetal forms.

KETOGENIC: A term used to describe amino acids that are metamobilised to acetoacetate and acetate.

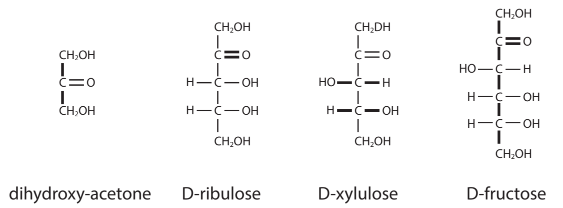
KETOL: An organic compound that has both an alcohol and a keto group. They are made by a condensation reaction between two ketones or by the oxidation of dihydric alcohols.

KETONE (ALKANONE): An organic compound containing a carbon atom connected to an oxygen atom by a double bond and to two other carbon atoms. Ketones are important in synthesising other compounds; and also important intermediates in cell metabolism. They are members of a family of organic compounds of general formula RCOR′, where R and R′ are organic radicals. Ketones may be made in various ways such as oxidation of secondary alcohols. The simplest ketone is acetone, an important industrial solvent.

KETONE BODIES: A mixture of water soluble ketone compounds: acetone, acetoacetic acid and β-hydroxybutyric acid, produced as by-products when fatty acids are broken down for energy in the liver and kidney. The concentration of ketone bodies in blood and urine increases in starvation, diabetes and pregnancy. In the brain, they are a vital source of energy during fasting.

KETONURIA (ACETONURIA; KETOACIDURIA): The presence in the urine of chemicals called ketone bodies. This may occur in diabetes mellitus, starvation and after persistent vomiting and results from the partial oxidation of fats.

KETOSE: One of the two major groups of monosaccharides, the other being the aldoses. It has a carbonyl (CO) group at a position other than on the terminal carbon atom, so forming a ketone group. It contains one ketone group per molecule. The simplest ketose is the three-carbon dihydroxy-acetone. Other ketoses include the five-carbon ribulose and xylulose and the six-carbon fructose and sorbose.



Types of Ketose

KETOSIS: 1. Accumulation in the blood and tissues of large quantities of the ketone bodies consisting of β-hydroxybutyric acid, acetoacetic acid and acetone. Because the first two are acids, this results in metabolic acidosis. Thus, the condition is often referred to as ketoacidosis. Ketone bodies are formed by ketogenesis when the liver glycogen stores are depleted, because of little intake of carbohydrates and the body has to break down fat for its energy. This results in the accumulation of ketone bodies in the blood. 2. A wasting disease in livestock caused by low level of glucose in the blood, which results from accumulation of waste products from the process of converting body fat to usable energy. The symptoms are a chronic lack of energy, depletion of fat reserves and a sudden drop in milk production.

KETTLE LAKE: A bowl-shaped, shallow, sediment-filled body of water formed by retreating glaciers or draining floodwaters.

KETYLS: Radical anions (or the corresponding salts) derived from ketones by addition of an electron.

KEY: 1. A biological system used for taxonomic identification. There are various types of keys but the two types which are commonly used for identification are the **dichotomous key** which is made up of contrasting and mutually exclusive pairs of statements; and the indented key. The morphological characteristics used for the identification may be qualitative or quantitative. 2. An electrical component that can open or close an electrical circuit. 3. A metal wedge or pin used to lock together two structural or mechanical components such as a shaft and a hub to prevent movement relative to each other. 4. A field in a database record that uniquely identifies that record. 5. A depressible on-off switch used for typing on a computer keyboard. Key switches are inherently digital devices that generate binary electric signal when depressed that can be readily interfaced with a microprocessor. 6. In cryptography, the sequence of symbols or characters that defines the makeup of an encoding mechanism.

KEY BED: A layer in a rock that has sufficiently distinctive characteristics such as coal beds and beds of volcanic ash showing an event that occurred quickly and over a widespread area. Key beds are traceable over a large horizontal and is used like an index fossil.

KEY CAP: The top part of a key on a keyboard that is pressed. It is placed on top of the electromechanical part or keyboard switch that moves down when pressed and springs back when released.

KEY FIELD: A unique identifier for a record in a file or database that identifies that record from all the other records. A database should always contain a key field. Examples are health insurance number, account number, product code and candidate number.

KEY PEST: A pest that occurs on a regular basis for a given crop.

KEY-TO-DISK SYSTEM (KEY-TO-TAPE SYSTEM): In computing, it is an obsolete system for data entry consisting of a processor with many terminals that enables large amounts of data to be entered at a keyboard, stored on a central a magnetic disk and transferred directly onto the computer.

KEYBOARD: An input device resembling a typewriter keyboard used to enter instructions and data into a computer. The keyboard layout is known as the **QWERTY design**, which gets its name from the first six letters across in the upper-left-hand corner of the keyboard.

KEYBOARD SHORTCUT: A key combination that performs a certain command, such as closing a window or saving a file. For example, pressing "Control-S" in a Windows program or "Command-S" on the Mac is the standard shortcut for saving an open document.

KEYLOGGER: A surveillance program that records the keystrokes on a computer by monitoring a user's input and keeping a log of all keys that are pressed. The log may be saved to a file or even sent to another machine over a network or the Internet.

KEYSTONE PREDATOR: A predatory species that helps maintain species richness in a community by reducing the density of populations of the best competitors so that populations of less competitive species are maintained.

KEYSTONE SPECIES: A species such as the beaver, sea otters or bison, whose impact on its community is disproportionately large relative to its abundance. They are important indicators of the environmental health of their habitats. An ecosystem may experience a dramatic shift if a keystone species is removed, even though that species was a small part of the ecosystem by measures of biomass or productivity. An example is elephants eating small trees so that the grassland is preserved or maintained.

K

KEYSTROKE: A stroke of a key, as on a computer keyboard or typing one character on a keyboard. Every time a key is hit, keystroke is performed.

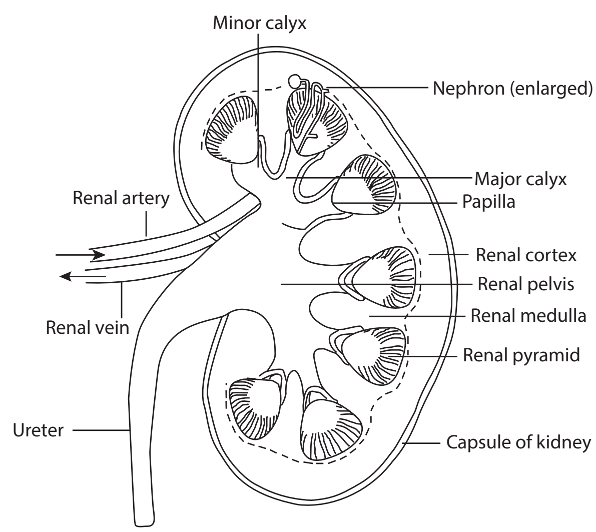
KEYSTROKE LOGGER (KEYLOGGER; SYSTEM MONITOR): A program or hardware device that captures every key depression on a specific computer's keyboard. As the user types, the device collects each keystroke and saves it as text in its own miniature hard drive. The device can be removed in order to access the information it has gathered. A software-based keystroke logger can also be a Trojan that is installed with the intent to steal passwords and confidential information. A keylogger program typically consists of two files that are installed in the same directory. One consists of a dynamic link library (DLL) file (which does all the recording) and the other is an executable file (.EXE) that installs the DLL file and triggers it to work. The keylogger program records each keystroke the user types and uploads the information over the Internet periodically to whoever installed the program.

KEYWORDS: Words or phrases that describe content. They can be used as metadata to describe images, text documents, database records and Web pages. Whenever you search for something using a search engine, you type keywords that tell the search engine what to search for. For example, if you are searching for houses, you may enter "houses" as keywords. Keywords can also describe the content of a Web page using the keyword meta tag. This tag is placed in the <head> section of a page's HTML and contains words that describe the content of the Web page.

KHACHIYAN ALGORITHM: The first polynomial time algorithm for linear programming based on the ellipsoidal method.

KID-SNATCHING: The practice of taking away a new-born kid from an animal such as a goat from its mother to prevent her from licking it and so passing on caprine arthritis- encephalitis (disease of goats involving synovial lined connective tissue, caused by a a member of the family *Retroviridae*, subfamily *Lentivirinae*).

KIDNEY: One of the pair of organs responsible for excretion of waste products, water regulation and maintaining the ionic composition of blood. It is the main organ used by vertebrates for excretion. Human kidneys are bean-shaped organs about 10 cm long located close to the spine dorsally in the body cavity. They consist of a number of smaller functional units called urinary tubules or nephrons. The nephrons filter the blood and cause wastes to be removed in the form of urine. The kidneys together with the bladder, two ureters and the single urethra, make up the body's urinary system. The functional unit of the kidney is the nephron.



Anatomy of the kidney

KIDNEY BEAN: A kidney-shaped bean, *Phaseolus vulgaris* is a herbaceous annual plant in the Fabaceae (legume or bean family) that is cultivated in many parts of the world for its beans used as a vegetable and noted for causing flatulence. Numerous cultivars have been developed, including string beans, stringless varieties such as the slender French haricot varieties and snap beans. It has the bush and twining varieties. The bush forms, which grow in erect leafy clumps, reach 1 m tall, while the twining forms, which need to be supported with poles or trellising, grow up to 4 m long. All varieties have trifoliate compound leaves with oval to rhombic leaflets, each up to 16 cm long, which are pubescent or covered with downy hair. The flowers, may be white, yellow, violet or red, in loose, open unbranched clusters (racemes) that are shorter than the leaves and develop into linear round to slightly flattened pods up to 15 cm long, each containing 4-6 beans. The beans are smooth, plump, kidney-shaped, up to 1.5 cm long. The beans may be red or white in colour and are often mottled in two or more colours. The bean is high in starch, protein and dietary fibre and is an excellent source of iron, potassium, selenium, molybdenum, thiamine, vitamin B6 and folate. It can be harvested and eaten immature, still in the edible pod or when mature, shelled and dried.

KIDNEY MACHINE (HAEMODIALYSIS): An artificial mechanical device which performs the functions of the kidneys, when they fail. It purifies the blood of a person with diseased kidneys. This is done through a process called **dialysis**. The process involves two needles being placed into the vascular access, one to remove the blood and the other to return cleansed blood to the body. The patient is connected by a tube to the dialysis machine through a vein in the arm. The blood is pumped from the body to a special filter called the dialyser, which is made of tiny capillaries.

KIDNEY TUBULE (RENAL TUBULE): Any nephron of the kidneys, which extends from the glomerulus in the Bowman's capsule that reabsorbs water and solute from the glomerular filtrate.

KIDNEYSTONE (NEPHROLITHIASIS): A hard small crystalline mineral material, formed within the kidney or urinary tract, consisting mainly of oxalates, phosphates and urates. Excessive amounts of protein in the diet can lead to high levels of calcium and increase the risk of developing kidney stones. Also, a decrease in urine volume, dehydration and certain medical conditions such as gout and those who take certain medications or supplements are at risk for developing kidney stones. Some kidney stones are small enough to pass through the urinary tract for elimination in the urine. Some may be large enough to obstruct the kidney and cause pain.

KIESELGUHR (DIATOMACEOUS EARTH; DIATOMITE): Silica containing mineral, a whitish powder that consists mainly of accumulated shells of diatoms. It has a particle size ranging from less than 1 micrometre to more than 1 millimetre, but typically 10 to 200 micrometres. It is used in fire proof cements, in the manufacture of dynamite, in insulating materials and as insecticide.

KIESERITE: A highly unstable magnesium sulfate powder with the chemical formula $MgSO_4 \cdot H_2O$, used as fertiliser where magnesium deficiency is evident, especially in light sandy soil.

KIGELIA: A genus of flowering plants in the family *Bignoniaceae*, comprising only one species, *Kigelia africana* also known as *Kigelia pinnata* and commonly known as sausage tree or cucumber tree. It is a medium-sized tree growing up to between 15 - 30 metres high with a dense spreading crown. It is easily recognised by the large sausage-shaped fruits hanging from its branches. The tree is found only in Africa on riverbanks, along streams and on floodplains and also in open woodland. It has a squat trunk and has a light brown, sometimes flaky bark and supports a dense rounded to spreading crown of leathery, slightly glossy foliage and is deciduous. The leaves are opposite and crowded at the ends of the branches, with 3-5 pairs of oblong leaflets and imparipinnate. The flowers are dark-red and open only at night and have strong unpleasant smell and are pollinated by bats and hawk-moths. The trumpet-shaped fruit is slender grey-green or grey-brown and hangs from a long stalk and is similar to grey sausages and can weigh up to 10 kg. The fruit pods are very fibrous with numerous hard seeds. *Kigelia africana*'s known chemicals include napthaquinones, fatty acids, courmarins, iridoids, caffeic acid, norviburtinal and sterols including sitosterol and stigmasterol and flavonoids. The unripe fruit is poisonous and if eaten, it is said to be a strong purgative and causes blisters in the mouth and on the skin. The ripe fruit is inedible, but the roasted or parched fruit or seeds respectively are placed in traditional grain beer for fermentation and to flavour it. It is also used as a medication against dysentery rheumatism and tape worms. Extracts of the tree are used as herbal relief for eczema and psoriasis. Also, traditional remedies are prepared from the crushed, dried or fresh fruits and used to treat ulcers, sores and syphilis. Beauty products and skin ointments are prepared from the fruit extracts.

KILL SWITCH: In computer science, it is a hardware or software that immediately terminates or shuts down a process or a device in an emergency. A kill switch can be activated remotely to stop a device such as a smartphone from being used by an unauthorised person or to prevent software piracy. It can also be used to shut down, for example, a stolen motor vehicle.

KILL FILE (KILLFILE): In computing, it is a file in a Usenet program specifying material such as names, posts, subjects or phrases that one does not wish to see when accessing a newsgroup. By entering such materials into the file, they are automatically deleted without being displayed.

KILLER CELL (CYTOTOXIC T CELL; KILLER T CELL): Either of two types of lymphocytes that destroy infected or cancerous host cells. Natural killer cells are distinct from both T cells and B cells in that they act without stimulation by specific antigen. They attack and destroy cancerous and virus infected cells that lack the normal class 1 histocompatibility proteins on their surface.

KILLER APPLICATION (KILLER APP): In computing, it is an application program or software so compelling or desirable that a potential user buys the hardware or computer that the program runs on just to run the application. An example is the first spreadsheet program, VisiCalc and Lotus 1-2-3.

KILLING AGE: In agriculture, it is the age at which an animal or bird is ready for slaughtering.

KILN: A furnace used for firing ceramics. The best known type is the rotary kiln used chiefly for cement, in which the burners are at the lower end of a slowly rotating cylinder, perhaps 100 metres long, which is at a slight angle to the horizontal.

KILO JOULE: 1. The SI unit of energy or work, which is equal to 1000 joules with the symbol kJ. 2. It approximates the energy needed to increase the temperature of 1 kilogram of water by 1 °C. 4.186 kilojoules (kJ) is equal to 1,000 calories or 1 kilocalorie or 1 Calorie (with an uppercase C).

KILO-: A prefix used in the metric system to denote 1000 times (10^3), with the symbol k. For example, 1000 volts = 1 kilovolts (kV), 2000 metres = 2 kilometres (km).

KILOBASE: A unit used at the molecular level for measuring distances along nucleic acid, chromosomes or genes, equal to 1000 bases.

KILOBYTE: The measurement of a computer memory size. One kilobyte is equal to 1024 bytes. Byte is an abbreviation for **binary term**, which is a unit of storage capable of holding a single character. One byte is equal to 8 bits. A bit is the smallest unit of information, which can be stored in a computer or a peripheral device, expressed as 0 or 1. Eight bits make a byte, the common measure of memory or storage capacity. Large amounts of memory are indicated in terms of kilobytes (1,024 bytes), megabytes (1,048,576 bytes) and gigabytes (1,073,741,824 bytes).

KILOCALORIE (KCAL): A non SI unit of heat measurement, which is equal to 1000 calories. It is defined as the amount of heat needed to raise the temperature of 1g of water by 1 °C at normal atmospheric pressure. It is often used to measure the energy content of food and the amount of energy burnt through exercising and other physical activities by an individual. The SI unit, joule is used as the unit of energy, although kcal is widely used. 1 kcal = 4.18 kJ and 1 kJ = 0.24 kcal.

KILOCYCLE: A former measure of frequency of a thousand cycles of any periodic phenomenon. It is equivalent to 1 kilohertz.

KILODALTON (kDa): A non-SI unit of mass, used to express atomic mass unit (amu) or molecular mass especially of large molecules, such as proteins and polysaccharides. Its symbol is u. A Dalton is exactly equal to one twelfth of the mass of an atom of carbon-12, the isotope of carbon with six protons and six neutrons in its nucleus, which is about 1.660×10^{-27} kg.

KILOGRAM: 1. The SI unit of mass denoted by kg, which is equivalent to the international standard of platinum-iridium cylinder kept at Sèvres near Paris. It is equal to 1000 grams. 2. A unit of force and weight, equal to the force that produces an acceleration of 9.80665 meters per second per second when acting on a mass of one kilogram.

KILOHERTZ: A unit of frequency denoted by kHz. 1 kHz = 1000 hertz (cycles per second).

KILOLITRE: A capacity measurement, which is equal to 1000 litres.

KILOMETRE: A unit of length denoted by km, which is equal to 1000 metres.

KILOPARSEC: A unit used for measuring large astronomical distances denoted by kpc. 1 kpc = 1,000 parsecs = 3,259 light-years.

KILOSTREAM LINK: In computing, it is a British Telecom-based service offering either data or voice transmission between sites in the UK. There are two types: KiloStream UK has transfer rates of 48 or 64 kilobits per second (kbps) for users that run low-speed applications such as remote printing facilities; and KiloStream N has transfer rates of 128 kbps, 25kbps or 512kbps for users that connect high-speed LANs, transmit high call volumes between sites or use video conferencing.

KILOTON: A unit for the power of nuclear explosion or warhead, which is equivalent to the explosion of 1000 tons of trinitrotoluene (TNT).

KILOTON WEAPON: A nuclear weapon with an explosive power equivalent to one thousand tons of trinitrotoluene (TNT).

KILOWATT-HOUR: The practical unit or Board of Trade Unit (BTU) of electrical energy, which is the commercial billing unit for electrical energy delivered to consumers. It is not a unit of power, but of energy. It is equivalent to a power consumption of 1000 watts for 1 hour or power in watts multiplied by time in hours. Its unit is kWh. $1\,\text{kWh} = 1\,\text{kW} \times 1\,\text{hr} = 1000\,\text{W} \times 60 \times 60 = 3.6\,\text{Mj}.$ Therefore, 1 unit of electric energy consumed = 1 kWh = 3.6 MJ.

KILOWATTS: A unit of power, which is equal to 1000 watts. Its symbol is kW.

KIMBALLTAG: A stock control device commonly used in cloth shops, consisting of a small punched card attached to items offered for sale.

KIMBERLITE: A rare igneous rock that occurs in the Earth's crust in vertical structures known as kimberlite pipes. It is found chiefly in South Africa and named after Kimberly, a town in that country. Often containing diamonds, it is the most important source of mined diamonds. It is composed of olivine and phlogopite mica, usually with calcite, serpentine and other minerals.

KIN SELECTION: Natural selection of genes that tend to cause the individuals bearing them to be altruistic to close relatives, even at a cost to their own survival and reproduction. For example, in an eusocial insect colony, sterile females act as workers to assist their mother in the production of other offspring. These relatives, therefore, have a higher probability of bearing identical copies of those same genes than do other members of the population. Thus kin selection for a gene that tends to cause beneficial characteristics such as an animal sharing food with close relatives will be spread throughout the population; on the other hand a gene that lowers the individual fitness of its carriers will be eliminated.

K

KINASE (PHOSPHOTRANSFERASE; PHOSPHOKINASE): An enzyme capable of transferring a phosphate group from high-energy donor molecules such as adenosine triphosphate (ATP) to acceptor organic molecules. This process is referred to as phosphorylation, which is the transfer of a phosphate group to an acceptor molecule, usually with ATP serving as the phosphate (phosphoryl) donor. Kinase activates the precursors of enzymes secreted in pancreatic juice. Often named after their substrates, kinases are used extensively to transmit signals and control complex processes in cells. An example is protein kinases which act and modify the activities of specific proteins.

KINDLE: In information technology, it is a portable e-reader used to download and read digital books, newspapers, magazines and other electronic publications. It also includes a built-in speaker and headphone jack for listening to audio books or background music. It was developed by Amazon.com.

KINDLING: 1. Material, such as dry sticks of wood that can easily ignite, used to start a fire. 2. Giving birth, as to kittens or rabbits. The litter may also be called a **kindle**.

KINEMATIC EQUATION (EQUATIONS OF MOTION): Any of four equations that apply to bodies moving linearly with uniform acceleration, a. These equations relate distance covered (s) to time taken (t) by the body: $v = u + at; s = \frac{1}{2}(u+v)t; s = ut + \frac{1}{2}at^2; v^2 = u^2 + 2as$, where u and v are the initial and final velocities of the body respectively, a the acceleration of the body, s the distance travelled by the body and t time used to cover the distance.

KINEMATIC VISCOSITY (COEFFICIENT OF KINEMATIC VISCOSITY): The ratio of the viscosity of a liquid to its density. Its unit is $m^2\,s^{-1}$.

KINEMATICS: The branch of mechanics concerned with the motions of objects without reference to the forces which act on the system. Kinematics is used in astrophysics to describe the motion of celestial bodies and systems and in mechanical engineering, robotics and biomechanics to describe the motion of systems composed of joined parts (multi-link systems) such as an engine, a robotic arm or the skeleton of the human body.

KINESIN: A motor protein, consisting of two heavy and two light chains. It is structurally similar to myosin. It uses chemical energy from adenosine triphosphate (ATP) to transport materials inside the cell by binding to microtubules and organelles. It is also involved in the assembly of the mitotic spindle and segregation of chromosomes, during their division.

KINESIS: The movement of a cell or organism in response to a stimulus such as light. The movement can be in any direction and its rate depends on the intensity of stimulation. The two main types of kineses include **orthokinesis**, in which the speed of movement of the individual is dependent upon the intensity of the stimulus; and **klinokinesis** in which the frequency or rate of turning is proportional to stimulus intensity.

KINETIC ACTIVITY FACTOR: A factor involving activity coefficients that appears as a multiplier in the rate equation.

KINETIC CONTROL: The term characterises conditions including reaction times that lead to reaction products in a reaction governed by the relative rates of the parallel (forward) reactions in which the products are formed, rather than by the respective overall equilibrium constants.

KINETIC CURRENT: A faradaic current that corresponds to the reduction or oxidation of an electroactive substance formed by a prior chemical reaction from another substance that is not electroactive and that is partially or entirely controlled by the rate of that reaction. The reaction may be heterogeneous, occurring at an electrode-solution interface (surface reaction) or it may be homogeneous, occurring at some distance from the interface (volume reaction).

KINETIC EFFECT: A chemical effect that depends on reaction rate rather than on thermodynamics. For example, diamond is thermodynamically less stable than graphite. Its apparent stability depends on the vanishingly slow rate at which it is converted. A reaction may be thermodynamically feasible (i.e has negative ΔH) but kinetically not feasible (i.e reaction is too slow). For example, the reaction $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O$ has standard enthalpy change of formation of $\Delta H_f = -286\,kJ\,mol^{-1}$, which suggests that thermodynamically, it is very feasible. However, this reaction takes about a billion years to occur making it kinetically not feasible.

KINETIC ENERGY: The energy possessed by an object due to its motion, which is a scalar quantity. The magnitude of the kinetic energy depends on both the mass and the velocity of the object, expressed as $E_k = ½mv^2$, where m = mass of the moving body and v = velocity of the moving body.

KINETIC EQUATIONS: Equations used in kinetic theory. A key application of kinetic equations is to calculate transport coefficients such as conductivity and viscosity in non-equilibrium statistical mechanics. In general, kinetic equations do not have exact solutions for interacting systems. An example of a kinetic equation is the Boltzmann Equation.

KINETIC METHOD: An analytical method in which the rate of a reaction or a related quantity is measured and utilised to determine concentrations.

KINETIC MODEL: A model which portrays the most common picture of the structure of matter. It is described by four statements: (i) all matter consists of particles; (ii) the particles are always moving; (iii) the particles tend to attract each other; and (iv) temperature relates to the mean energy of the particles.

KINETIC RESOLUTION: The achievement of partial or complete resolution by virtue of unequal rates of reaction of the enantiomers in a racemate with a chiral agent (reagent, catalyst, solvent, etc.).

KINETIC TEMPERATURE: The temperature of a gas defined in terms of the average kinetic energy of its atoms or molecules. It is given by: $T = \frac{2}{3}\frac{1}{k} avg\left(\frac{1}{2}mv^2\right)$, where T is the kinetic temperature, k is Boltzmann's constant, m is the particle mass and v is the particle velocity. In thermal equilibrium, values of kinetic temperature correspond well with those of other measures of temperature, such as effective temperature; otherwise, they may bear little resemblance.

KINETIC THEORY OF MATTER: A theory describing the physical properties of matter in terms of the behaviour, especially the movement of its component atoms or molecules. The theory states that all matter are made of small particles (atoms and molecules) that have spaces between them and are in constant motion and that the extent to which movement can occur depends on the temperature. It explains the behaviour of solids, liquids and gases by assuming that they are made up of particles that are always moving.

KINETIC THEORY OF GASES: A theory that explains the behaviour of gases as follows: (i) Any gas is composed of very large number of molecules and the collision of the molecules is perfectly elastic; (ii) the forces between gas molecules are negligible; (iii) the gas molecules move randomly in straight line; (iv) they collide with each other and the walls of the container; and (v) the volume of the gas is negligible compared to the volume of the container. A gas corresponding to these assumptions is called an **ideal gas**.

KINETIC VISCOSITY: The ratio of the viscosity of a liquid to its density. It is denoted by v. Its SI unit is m^2s^{-1}.

KINETIC-MOLECULAR THEORY: A theory, that attempts to explain macroscopic observations of gases in microscopic observations or molecular terms.

KINETICS: 1. The branch of physical chemistry concerned with measuring and studying the rates of chemical reactions, i.e., change in concentration of a reactant or a product with respect to time. 2. The branch of physics dealing with the action of forces producing or changing the motion of a body. Its aim is to determine the mechanism of reactions.

KINETIN (6-FURFURYLAMINOPURINE): A synthetic cytokinin, which was first isolated from DNA extracts of animal origin but has now also been synthesised commercially.

KINETOCHORE: A protein structure on chromosomes by which the microtubules of the spindle is attached to the centromere of a chromosome during cell division to pull the chromosomes apart. In higher organisms it consists of protein and RNA arranged in a layer closely opposed to the chromatin. The kinetochore contains two regions: an inner kinetochore, which is tightly associated with the centromere DNA; and an outer kinetochore, which interacts with microtubules. It is the site of chromosome tubule attachment.

KINETOSOME (BASAL BODY; BLEPHAROPLAST): A cylindrical structure in the cytoplasm from which a flagellum or cilium arises in motile eukaryotic cells. It is composed of a cylinder of nine microtubules linked by fine crossbridges. It transfers adenosine triphosphate (ATP) and possibly other materials to the fibres in the shaft of the flagellum or cilium. Striated fibrous extensions, termed rootlet fibres, may arise from the basal body and project backwards into the cytoplasm.

KINGDOM: The highest taxonomic rank used in biological classification of all sorts of living organism. In the three-domain system, it is the rank below domain. The major kingdoms are Animalia (animals), Plantae (plants), Fungi (fungi), Prototista (single-celled eukaryotes), Prokaryota (single-celled prokaryotic bateria and cyanobacteria) and Monera (prokaryotes). Kingdoms are divided into smaller groups called phyla in zoology or divisions in botany.

KININS (KININ-KALLIKREIN SYSTEM; KININ SYSTEM): 1. A polypeptide produced from kininogen by kallikrein that causes dilation in blood vessels, contraction of smooth muscle and alteration of vascular permeability. Kinins are broken down by kininases. They act on phospholipase and increase arachidonic acid release and thus prostaglandin production. They are also involved in blood pressure control, coagulation and pain. 2. Also called **cytokinin**, it refers to any of a group of plant hormones which affect growth and are used to slow down the aging process in plants, encourage seeds to germinate and plants to flower and are involved in plant responses to drought and waterlogging. They are used commercially to produce seedless grapes, to stimulate germination of barley in brewing and to prolong the life of green-leaf vegetables. Kinins work in conjunction with auxins to promote swelling and division in the plant cells producing lateral buds. In animals, kinins are peptides released by enzyme action which contraction smooth muscle and elevate blood pressure, thus playing an important role in inflammation and shock. Kinins, such as zeatin, are derived from adenine.

KINOCILIUM: The single cilium of a hair cell located in the sensory epithelium of the vertebrate inner ear that protrudes much further than the other relatively short stereocilla. It is involved in the senses of movement and hearing.

KIOSK: In information technology, any physical structure containing a computer set up to display information such as to direct people in a public place. Users can navigate the display using keyboards or touch screens, but prevents users from accessing the computer's operating system. A kiosk in a museum might show an interactive multimedia display or one in a library might give readers access to catalogues.

KIP: The untanned hide of a small or young animal, such as a calf.

KIPP'S APPARATUS: A laboratory apparatus for making small volumes of gases by the reaction of a solid with a liquid without heating. An example is the reaction of hydrochloric acid with iron sulfide to give hydrogen sulfide.

KIRCHOFF, GUSTAV ROBERT (1824-87): German physicist, born in Königsberg, Prussia, now Kaliningrad in Russia. He was professor of physics at the universities of Breslau, Heidelberg and Berlin. With the German chemist Robert Wilhelm Bunsen, they developed the modern spectroscope for chemical analysis, which led them to discover the elements caesium in 1860 and rubidium in 1861. In 1845, he formulated Kirchoff's laws concerning electric circuits, which allows the calculation of currents, voltages and resistances of electrical networks and generalised the equations describing current flow to three dimensions.

KIRCHOFF'S LAW OF THERMAL RADIATION (KIRCHOFF'S PRINCIPLE): The law proposed by Gustav Kirchoff in 1859, which states that the ratio of the emissivity of an object to the absorptivity of the same object is the same for all bodies, depending on frequency and temperature alone and is equal to the emissivity of a blackbody.

KIRCHOFF'S LAWS: A pair of laws governing the flow of current and voltage in an electrical circuit, named after German physicist, Gustav Robert Kirchoff (1824-87). The first law, also referred to as **Kirchoff's current law**, the **point rule** or the **junction rule** states that in a closed electrical circuit, the algebraic sum of currents at a junction is zero ($\sum I = 0$) or in a closed electrical circuit, the sum of currents entering a junction equals the sum of currents leaving the junction ($I = I_1 + I_2$), therefore $I - (I_1 + I_2) = 0$. The first law deals with conservation of charges and implies that charges are conserved. The second law, also referred to as **Kirchoff's**

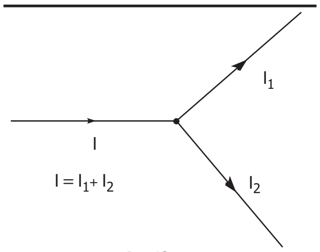


Fig. i) Kirchoff's First Law (Kirchoff's Current Law)

voltage law or the **loop rule** states that the algebraic sum of the electromotive forces within any closed circuit is equal to the sum of the potential difference in the various portions of the circuit or at any junction of paths or node: $\sum E = \sum IR$. In other words, the algebraic sum of all the electromotive forces and the voltage drops is equal to zero. Referring to the diagram, loop

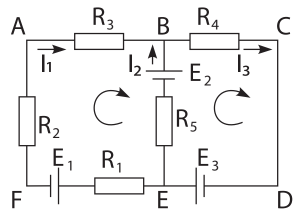


Fig ii.) ***Kirchoff's Second Law (Kirchoff's voltage law)***

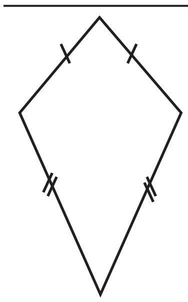
$$ABEFA = -E_3 - E_1 = -I_2R_5 - I_1R_1 - I_1R_2 - I_1R_3 \Rightarrow$$

$E_1 + E_2 = I_2R_5 + I_1R_1 + I_1R_2 + I_1R_3$. The second law is a statement of energy conservation, because any charge that starts and ends up at the same point with the same velocity must have gained as much energy as it lost.

KISH: In metallurgy, it is a variety of graphite containing a large amount of carbon that is formed on the surface of molten pig iron or cast iron as it cools in iron-smelting furnaces.

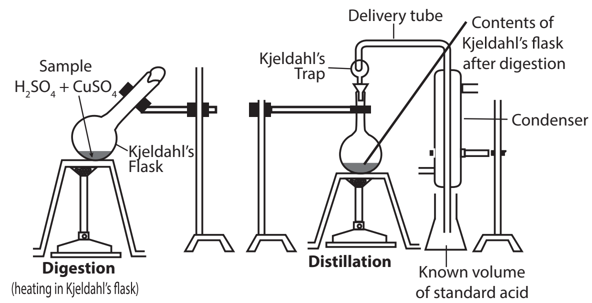
KISS OF LIFE (ARTIFICIAL RESPIRATION): Any procedure used to supply air or oxygen to the lungs of a suffocating person to restore normal breathing or a method of resuscitating a person who suffers from asphyxia or lack of oxygen. It assists or stimulates respiration, a metabolic process to have the overall exchange of gases in the body by pulmonary ventilation, external respiration and internal respiration. The procedure takes many forms, but generally entails providing air for a person who is not breathing or is not making sufficient respiratory effort on their own. However, it must be used on a patient with a beating heart or as part of cardiopulmonary resuscitation to achieve the internal respiration. Some common methods used include the following: (i) **Mouth to mouth**, with the rescuer making a seal between his or her mouth and the patient's mouth and blowing air into the patient's body; (ii) **Mouth to nose**, in which the rescuer may need to form a seal with the patient's nose; and (iii) **Mechanical resuscitator**, which uses an electric unit designed to breathe for patients. All methods, however require good airway management to perform, which ensures that the method is effective.

KITE: 1. A geometric figure, which forms a quadrilateral or a four sided figure with two pairs of adjacent equal sides. The angles are equal where the pairs meet. The diagonals meet at a right angle and the longer diagonal is the axis of symmetry. 2. An unpowered, heavier-than-air flying device held to the Earth by a line. The wind resistance causes the air pressure under the kite to be greater than the air pressure above it, making it rise into the air and fly. 3. Any of various predatory birds of the hawk family Accipitridae, subfamily Aegypiinae having a long, elongated forked tail and long pointed wings. It steals prey from other birds of prey.



Kite (a geometric figure)

KJELDAHL'S METHOD: A technique for the quantitative determination of nitrogen in an organic compound, developed by Johan Kjeldahl in 1883. The compound is heated with concentrated sulfuric acid to convert any nitrogen to ammonium sulfate using copper(II) sulfate as a catalyst. An alkali is added and the mixture is heated to distill off ammonia. This is passed through a known volume of standard acid solution. The amount of ammonia is found by estimating the amount of unreacted acid by titration. The amount of nitrogen in the original specimen can then be calculated.



Apparatus for estimating nitrogen by Kjeldahl's method

KLEIBER'S LAW: The relationship in which metabolic rate (R) of an organism varies with its body mass (M) raised to the power ¾, expressed as R x $M^{3/4}$. For example, an animal, having a mass 100 times that of a smaller animal, will have a metabolism roughly 31 times greater than that of the smaller animal. In plants, the exponent is close to 1. This law has been shown to apply to organisms ranging in size from bacteria to whales as well as plants.

KLEIN BOTTLE: A closed surface with only one-side and no interior that produces two Mobius strips if it is cut in half lengthwise. It can be formed by inserting the narrower end of an open tapered tube through the wider end and then sealing the ends to one another.

KLEIN FOUR-GROUP: The smallest non-cyclic group consisting of four permutations $e = (1), a = (12)(34), b = (13(24), c = (14)(23).$

KLEIN-GORDON EQUATION (KLEIN-FOCK-GORDON EQUATION; KLEIN-GORDON-FOCK EQUATION): An equation in relativity quantum mechanics for spin-zero particles. It describes a spinless particle which is consistent with the special theory of relativity.

KLEIN, CHRISTIAN FELIX (1849 -1925): German mathematician, known for his contributions in group theory, complex analysis, non-Euclidean geometry and also introduced the Erlangen Programme.

KLEPTOMANIA: A behavioural disorder characterised by a great impulse to steal or possess items that usually have little value for which one has no need. Sometimes the sufferer has no memory of the theft.

KLEPTOPARASITISM: A type of obtaining food in which an organism steals prey or food from another organism. A **kleptoparasite** is an animal that obtains its food by this means. Mammals such as some hyena and jackals; birds such as skuas and frigatebirds; and insects such as cuckoo bees and cuckoo wasps of the family Chrysididae practise this behaviour. When the kleptoparasite steals food from members of other species, it is called **interspecific kleptoparasitism**; if it is obtained from members of its own species, it is called **intraspecific kleptoparasitism**. When a kleptoparasite derives its entire energy intake from stealing food from other animals, it is called **obligate kleptoparasitism**; if it has alternative sources of obtaining food, it is called **facultative kleptoparasitism**.

KLINEFELTER'S SYNDROME (XXY SYNDROME): A genetic disorder affecting men in which an individual gains an extra X chromosome, so that the usual karyotype of XY is changed to XXY. A person with Kleinfelter's syndrome has male features with firm, small testes which do not produce sperm and may have enlarged breasts and buttocks and very long legs. Testosterone is low and pituitary reproductive hormones high. All patients with Kleinfelter's disorder are treated with androgens.

KLINOSTAT (CLINOSTAT): Laboratory equipment used in experiments to test the influence of gravity on the growth movements of plants. It consists of a motor that slowly rotates a drum inside which seedlings are attached. This allows any single part of the seedlings from receiving uninterrupted gravitational stimulation thereby resulting in horizontal growth of the seedlings.

KLYSTRON: An evacuated electron tube that generates or amplifies microwaves by velocity modulation. It is used as amplifiers at microwave and radio frequencies to produce both low-power reference signals for superheterodyne radar receivers and to produce high-power carrier waves for communications and the driving force for modern particle accelerators. There are several types. In the simple two-cavity klystron, a beam of high-energy electrons from an electron gun is passed through a resonant cavity, where it interacts with high-frequency radio waves. This microwave energy modulates the velocities of the electrons in the beam, which then enters a drift space where the faster electrons overtake the slower ones to form bunches. The bunched beam now has an alternating component, which is transferred to an output cavity and then to an output waveguide.

KNACKER: A person who slaughters casualty animals, particularly horses.

KNAPSACK PROBLEM: The problem of determining which numbers from a given collection of numbers have been added together to yield a specific sum. Mathematically, it is the integer problem of maximising $\sum_{k=1}^{n} c_k x_k$ subject to $\sum_{k=0}^{n} w_k x_k \leq k.$ where the variables x_i; are non-negative integers. The problem is analogous to trying to maximise the value of goods packed into a knapsack. It is used in cryptography to encipher and sometimes decipher messages.

KNAPSACK SPRAYER: A spraying apparatus consisting of a pressurised chemical tank, a pump, filters, pressure gauge and a hand-held non-drip nozzle that is carried on the back. In agriculture, it is used to apply herbicides, pesticides, and fertilisers on crops. It can also be used to control small-scale fire.

KNEE: 1. The middle joint of the leg where it bends and the largest and most complex joint in the human body. It joins the thigh with the leg and articulates between the femur and tibia and also between the femur and patella. In humans, the knee supports nearly the whole weight of the body and is most vulnerable to both acute injury and the development of osteoarthritis. 2. In botany, it is any of the hollow rounded protuberances that project upwards from the roots of the swamp cypress. It is thought to aid respiration in waterlogged soil.

KNEECAP: 1. Also known as **patella** or **kneepan**, it is a flat, triangular bone at the front of the knee, which articulates with the femur and protects the knee joint. It is the largest sesamoid bone (bone that develops inside a tendon) in the body. 2. A felt protector for the knee of horses.

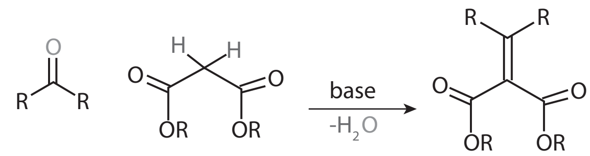
KNIFE APPLICATION: Injection of fertiliser into soil in gaseous form.

KNOCKIN: A technique in which a gene is inserted into the genome of a cell, cell line or organism. It can be induced to switch on only in certain tissues or at certain stages of development, thus mimicking the normal behaviour of genes.

KNOCKING (KNOCK; DETONATION; SPARK KNOCK; PINGING; PINKING): The explosive burning of fuel in a petrol engine causing loss of efficiency and damage or a phenomenon that occurs when unburnt fuel:air mixture explodes in the combustion chamber before being ignited by the spark. It is caused by rapid and uncontrolled combustion in the combustion chamber, ahead of the flame front and it occurs when the comprehension ratio of the engine is too high for the fuel being used.

KNOCKOUT: A technique for inactivating a particular gene or genes within an organism or cell. Both single-celled and multicellular organisms can be genetically engineered so that a normal gene is replaced with a defective homologous gene. Experiments with laboratory organisms treated in this way reveal how defects in particular genes can affect the development and life of the organism. Knockout mice are engineered by transfected embryos that are heterozygous for the defective gene.

KNOEVENAGEL REACTION: A condensation reaction between a carbonyl compound (aldehyde or ketone) and an active methylene compound (RCH_2R^1), where R and R^1 are acid, ester or cyanogen functional group with the elimination of water to form α,β-unsaturated compounds. They are usually catalysed by a base.



Knoevenagel Reaction

KNOT: 1. A unit of speed used in sea and air navigation, based on the length of one minute of arc of a great circle. It is equivalent to 1,852 metres per hour. 2. A closed curve in three dimensions. The two simplest nontrivial knots are the trefoil knot, which has three crossings; and the figure-eight knot also called Listing's knot, has four crossings. The trefoil knot can be obtained by joining together the two loose ends of a common overhand knot, resulting in a knotted loop.

KNOT GRASS: 1. A common widely distributed herb, *Polygonum aviculare* with jointed stems and small greenish flowers. It grows along irrigation and drainage ditches, canals, marshes, moist grassland, stream banks, moist disturbed areas, roadsides, rice fields, irrigated perennial crops orchards and vineyards. Its seeds and leaves serve as forage for numerous animals and birds. It is also used for controlling erosion in ditches and canal banks. Its spreading nature makes it problematic for crops. 2. *Acronicta rumicis*, a moth of the family Noctuidae found in Europe, North Africa and Asia, which are attracted to light. Its wingspan is 34-40 mm. The larvae feed on a range of herbaceous plants.

KNOT THEORY: A mathematical theory of closed curves in three-dimensional space. It is applied to the study of the properties of polymers such as DNA which is a polymer of biological importance and the statistical mechanics of certain models of phase transitions.

KNOWBOT: Artificial computer program that automates the search for animation.

KNOWLEDGE MANAGEMENT: A process by which an organisation gathers, organises, shares and analyses its knowledge. Data entered via a keyboard, a scanner or voice input can be catalogued, indexed, filtered and linked. The information can then be refined, for example through data mining, in preparation for dissemination.

KNOWLEDGE-BASED SYSTEM (KBS): A computer program that uses an encoding of human knowledge to solve complex problems. It was discovered during research into artificial intelligence that adding heuristics (rules of thumb) enabled programs to tackle problems that were otherwise difficult to solve by the usual techniques of computer science.

KNUDSEN FLOW: A model of the flow of gas in a container at a very low density, named after Martin Hans Christian Knudsen (1871-1949). For a gas passing through small holes in a thin wall in the Knudsen flow regime, the number of molecules that pass through a hole is proportional to the pressure of the gas and inversely proportional to its molecular weight. Knudsen flow can be used to do a partial separation of a mixture of gases if the components have different molecular weights. Knudsen flow uses porous membranes to separate isotopic mixtures such as uranium. In hydrogen production, it is used to separate hydrogen from the gaseous product mixture created when water is heated at high temperatures using solar or other energy sources.

KOCH CURVE: The fractal curve of Hausdorff dimension ln4/lnS, generated by a replacement rule that at each step, the middle third of each line segment must be replaced with two sides of a right triangle having sides of length equal to the replaced segment.

KOCH, ROBERT (1843-1910): German physician and medical bacteriologist, who demonstrated conclusively in 1876 that the serious infectious anthrax disease of domestic animals was caused

K

by a bacterium. His series of experiments in the study of the cause of anthrax was the first application experimentally of the principles, which had been laid down 36 years before by Koch's professor of anatomy, J. Henle, at the University of Göttingen. These principles are now generally known as Koch's postulates. Eminently, Koch discovered in 1881 that tuberculosis is caused by a bacterium and studies on the disease constituted a large part of his contribution to bacterial pathogens. He visited several countries including India, Zambia and India to solve raging epidemics. He was awarded the Noble Peace Prize for Medicine in 1895.

KOCH'S POSTULATES: Criteria which establish the causal relationship between a specific microorganism and a specific disease. These are: (i) the microorganism must be present in every case of the disease; (ii) the microorganism must be isolated from the diseased host and grown in culture; (iii) the specific disease must be reproduced when a pure culture of the microorganism is inoculated into a healthy susceptible host; (iv) the microorganism must be recovered once again from the experimentally infected host.

KOHLRAUSCH EQUATION: An equation that describes the molar conductivities of strong electrolytes at low concentration. It was named after Friedrich Wilhelm Georg Kohlrausch (1840-1910), a German physicist. The equation is given by: $\Lambda_m = \Lambda_m^0 - KC^{\frac{1}{2}}$, where Λ_m = the molar conductivity, Λ_m^0 = the limiting conductivity, K is a coefficient related to the stoichiometry of the electrolyte and C is the concentration of the electrolyte.

KOHLRAUSCH'S LAW: The law states that the conductivity of the solution of a salt dissolved in water is the sum of two values. One depends on the positive ions and the other on the negative ions. It was devised by Friedrich Wilhelm Georg Kohlrausch (1840 - 1910), a German physicist, who investigated the conductive properties of electrolytes.

KOLA: Two evergreen trees (*Cola acuminata* and *C. nitida*) of the cocoa family (*Sterculiaceae*), native to west Africa, which produce bitter edible nuts known as cola nuts that contain caffeine. The nut has been used in medicines and in soft drinks. Traditionally, they are chewed to diminish sensations of hunger and fatigue, to aid digestion and to combat intoxication, hangover and diarrhoea.

KOLMOGOROV COMPLEXITY: In information theory, it is the measure of the descriptive complexity of an object. It is related to, but different from, computational complexity, which measures the time or space required for a computation.

KOLMOGOROV-SMIRNOFF TEST: In statistics, the goodness of fit test of which the test statistics is $\max_{i=1\ldots n}\left|\frac{i}{n} - F(X_i)\right|$, where F is the hypothesised distribution and the X_i are in order. This test statistic is independent of F when the hypothesis is true.

KOLMOGOROV'S THREE SERIES THEOREM: The result that a series of independent random variables, $\{f_n\}$, converges pointwise (almost everywhere) if and only if three numerical series converge. When the variables are uniformly bounded, this reduces to verifying the convergence of the sums of the expectations, $E(f_n)$ and of the variances, $\sigma(f_n)$. It was named after Russian mathematician Andrey Kolmogorov (1903-87).

KOMODO DRAGON (KOMODO MONITOR): A large lizard (*Varanus komodoensis*) belonging to the family Varanidae and found in the island of Komodo and surrounding islands in Indonesia. It is the world's largest lizard, growing up to about 3 metres and weighing 70 kg.

KONIG'S THEOREM: The result that the number of edges in a maximum matching of a bipartite graph equals the number of vertices in a minimal covering of the graph, that is, in a subset K of vertices such that each edge has a vertex in K.

KONIGSBERG BRIDGE PROBLEM: A mathematical problem, set by Euler, a Swiss, in a town called Konigsberg. It is a problem of walking across seven bridges connecting four landmasses in a specified manner exactly once and returning to the starting point (an Eulerian circuit of the town). It was shown to be impossible by Euler's demonstration that an Eulerian circuit exists in a connected graph if and only if each vertex has even degree. It is the original problem which gave rise to the graph theory.

KOROVKIN THEOREMS: A class of theorems regarding uniform approximation of which the most basic is that if a sequence $\{L_n\}$ of non-negative linear operators on $C[a,b]$ is such that $L_n(x^k)$ converges uniformly to x^k for k = 0, 1, 2, then $L_n(f)$ converges uniformly to f for all continuous functions.

KORSAKOFF'S SYNDROME: A chronic memory disorder in which the brain fails to cope with new information, due to brain damage caused by severe deficiency of thiamine (vitamin B-1); distant memory is however preserved. It is most commonly caused by chronic alcoholism.

KOVALEVSKAYA, SOFYA (1850-1891): Russian mathematician who worked in Germany and Sweden and made important contributions in differential equations.

KOVAR: A trade name for an alloy of iron, nickel and cobalt with expansivity similar to that of glass in order to allow direct mechanical connections over a range of temperatures. It is used for making glass-to-metal seals, especially in circumstances in which a temperature variation can be expected, such as vacuum tubes, x-ray and microwave tubes and some light bulbs.

kph (km/h): Abbreviation for kilometres per hour.

KRAMERS' DEGENERACY THEOREM: It states that the energy levels of systems with an odd number of electrons remain at least doubly degenerate in the presence of purely electric field. Kramers degeneracy can be removed by placing the system in an external magnetic field. The theorem holds even in the presence of crystal fields and spin-orbit coupling.

KRAMERS' LAW: A formula derived in 1923 by the Dutch physicist Henrik Kramers (1894–1952), for the opacity of material inside a star. It states that, at high temperature, stellar opacity is proportional to the density divided by the temperature to the power of 3.5.

KRANZ ANATOMY: The arrangement of palisade mesophyll cells consisting of a sheath of tightly packed cells around the vascular bundle of C4 plants such as corn and sugar cane, giving a more rapid export of photosynthetic products from the leaf. This contributes to relatively faster growth in abundant light.

KREB'S CYCLE (CITRIC ACID CYCLE; TRICARBOXYLIC ACID CYCLE, TCA CYCLE): The final part of the chain of biochemical reactions essential for the oxidative metabolism of glucose and other simple sugars to form carbon dioxide, water and energy. The reaction in the Kreb's cycle is cyclical since the end product, oxaloacetic acid is also an initial compound for the cycle of reactions.

KREIN-MILMAN THEOREM: The theorem stating that a compact convex subset of a locally convex space is the closed convex hull of its extreme points.

KROLL PROCESS: A pyrometallurgical industrial process for producing certain metals such as metallic titanium and zirconium by reducing the chloride with magnesium metal.

KRONECKER DELTA: A function of two variables i and j that equals 1 when $i=j$ and equals 0 when the variables have different values: Kronecker delta, δ_{jk} = 1 if $j = k$ and 0 if $j \neq k$.

KRONECKER'S LEMMA: If x_1, x_2, ... and $0 < a_1 < a_2 < \ldots$ is a sequence of real numbers such that a_n increases to infinity as $n \to \infty$, and suppose that the sum, $\sum_{n-1}^{\infty}\frac{x_n}{a_n}$ converges to a finite limit, then $\frac{1}{a_n}\sum_{k=1}^{n}x_k \to 0$ as $n \to \infty$.

KRUMMHOLZ (ELFIN FOREST): A forest of short and stunted trees found in the zone just above the timber line on tropical mountains. Continual exposure to fierce freezing winds causes vegetation to become stunted and deformed. Some trees showing krumholtz formation include balsam fir, red spruce, subalpine larch and lodgepole pine.

KRUSKAL-WALLIS TEST: A distribution free method that is analogue of the analysis of variance of a one-way design. It tests whether the groups to be compared have the same population median. The test statistic is derived by ranking all the N observations from L to N regardless of which group they are in and then calculating, $H = \frac{12\sum_{i=1}^{k} n_j(\bar{R}_i - \bar{R})^2}{N(N-1)}$, where n_j is the number of observations in group i, $\bar{R}_i$ is the mean of their ranks, $\bar{R}$ is the average of all the ranks, given explicitly by $\frac{N+1}{2}$. When the null hypothesis is true, the test statistic has a chi-squared distribution with k-1 degrees of freedom.

KRYPTOL: Mixture of loose granule graphite, carborundum and clay with a high electrical resistance, used in electric furnaces.

KRYPTON: A colourless, odourless, gaseous, non-metallic element with the chemical symbol Kr, atomic number 36, atomic weight 83.80, melting point -156.6 °C (116.55 K), boiling point -152.30 °C (120.85 K) and density 3.73 grams per litre (0 °C). It belongs to the group of inert gases (Group 18, VIIIA of the Periodic Table). It occurs in trace amounts in the atmosphere and in rocks and extracted by fractional distillation of liquid air. It is used in luminescent tubes, flash lamps, incandescent light bulbs, lasers and some types are used in medicine to check lung function.

KUDOS: A social media platform designed for young people between the ages eight and thirteen that, besides sharing photos, comments and reactions online, teaches young users how to communicate and stay safe online. Kudos does not offer advertisements and parents are informed of their children's activities.

KUHN-TUCKER CONDITIONS (KARUSH-KUHN-TUCKER THEOREM): A result extending the Lagrange method of multipliers to constraints by equalities and inequalities. It ensures that, given an appropriate constraint qualification, any point x_0, that minimises $f(x)$ subject to $g_1(x) \le 0, ..., g_n(x) \le 0, g_{1+n}(x) = 0, ..., g_{m+n} = 0$ will occur at a stationary point of the Lagrangian $L(x, \lambda) = f(x) + \sum \lambda_i g_i$, where the summation is over the binding constraints with $g_k(x_0) = 0$ and where the multipliers corresponding to inequality constraints are nonnegative.

KUIPER BELT (EDGEWORTH–KUIPER BELT): A disc-shaped region of small icy objects made up of ice, dust and rock, which orbit the Sun beyond the orbits of Neptune and Pluto. It was named after Gerald Peter Kuiper (1905-1973), a Dutch-born U.S. astronomer. The Kuiper belt is similar to the asteroid belt, but about 20 times wider and 20-200 times as massive. The belt is home to at least three dwarf planets: Pluto, Haumea and Makemake. Some of the Solar System's moons, such as Neptune's Triton and Saturn's Phoebe, are also believed to have originated in the region.

KUNDT'S TUBE: An acoustic apparatus designed by August Kundt (1839-94) for measuring the speed of sound in various fluids. It consists of a closed glass tube into which a dry powder like lycopodium has been sprinkled. The original device was a metal rod clamped at its centre with a piston at one end which is inserted into the tube. When the rod is stroked, the piston enters the tube, to produce sound waves. Adjusting the position of the piston in the tube so that the gas column becomes a whole number of half wavelengths long, the dust will be disturbed by the resulting stationary waves forming a series of striations, enabling distances between nodes to be measured. The vibrating rod can be replaced by a small loud speaker fed by an oscillator.

KUPFERNICKEL (NICCOLITE; NICKELINE): Natural nickel arsenide (NiAS) in hexagonal crystalline form containing 43.9% nickel and 56.1% arsenic. It is a copper-coloured mineral, which is an important ore of nickel.

KUPFFER CELLS (MONONUCLEAR PHAGOCYTE SYSTEM): Specialised cell macrophages located in the liver lining the walls of the sinusoids that dispose of worn-out bloods cells, bacteria and particulate matter from the bloodstream. It was named after Karl Wilhelm von Kupffer (1829-1902), a Bavarian anatomist.

KURATOWSKI–ZORN LEMMA (ZORN'S LEMMA): In mathematics, it is a theorem of set theory that in an ordered set in which every chain has an upper bound, there is an element in the set such that the set contains no element greater than the specified given element. It is named after the German-born American mathematician, Max August Zorn (1906-1993) and Kazimierz Kuratowski (1896-1980), Polish mathematician.

KURTOSIS: In statistics, it is a measure of the concentration of a distribution about its mean, especially $B_2 = \frac{m_4}{(m_2)^2}$, where m_2 and m_4 are, respectively, the second and fourth moments of the distribubon about the mean. A normal distributlon or any other with $B_2 = 3$ is called **mesokurtic**; $B_2 < 3$ is **platykurtic** and if $B_2 > 3$ it is **letokurtic**.

KVM SWITCH (KEYBOARD, VIDEO, MOUSE SWITCH): A hardware or software device that can use a single keyboard, video monitor and mouse to control multiple computers at a time. Examples are Hardware KVM Switch, USB Hub Based KVM (Enumerated KVM switch), Emulated USB KVM and DDM USB KVM.

kW: Abbreviation for kilowatt, a unit of electric power.

KWASHIORKOR: A tropical disease of children caused by eating food that does not contain enough protein or a severe protein deficiency in children under five years, resulting in retarded growth and a swollen abdomen, swelling of the hands, feet and face and the appearance of discoloured blotches on various parts of the body.

KYANITE: A bluish aluminosilicate mineral, with the chemical formula Al_2SiO_5, found as thin-bladed crystals or in masses. It is derived from metamorphic rocks and used for the manufacture of highly refractory porcelains such as those used for spark plugs and as gems.

KYLOES (HIGHLAND CATTLE): A breed of small long-horned shaggy beef cattle, which originated from Scotland.

KYMOGRAPH: A device for recording variations in motion or pressure such as blood pressure, consisting typically of a rotating drum and a stylus. It was invented by German physiologist Carl Ludwig in the 1840s.

KYOTO MECHANISMS (FLEXIBILITY MECHANISMS): Economic mechanisms based on market principles that countries to the Kyoto Protocol can use to reduce greenhouse gas emission in a cost-effective way.

KYOTO PROTOCOL (KYOTO ACCORD): An amendment to the United Nations Framework Convention on Climate Change in which signatory countries agree to reduce carbon dioxide emissions and the presence of greenhouse gases. It was adopted in Kyoto, Japan on 11 December 1997 and entered into force on 16 February 2005.

L

L (l): Symbol used to denote the number fifty in the Roman numerals. The upper or lower case letters can be used.

L-DOPA (LEVODOPA): The levorotatory form of dopa, normally produced by the body and converted by an enzyme to dopamine in the brain. It is used to treat Parkinson's disease.

L-FORM: A defective bacterial cell, of indefinite or spherical shape, in which the cell wall is defective or absent. L forms may develop spontaneously or in response to a variety of stimuli, such as thermal or osmotic shock or the presence of antibiotics that inhibit cell-wall synthesis. They may revert to their original condition on removal of the stimulus or they may reproduce as L forms. On solid media, L forms have a characteristic colonial appearance resembling that of mycoplasmas with which they were originally confused.

L-FRUCTOSE (LAEVULOSE; FRUCTOSE; FRUIT SUGAR; 1,3,4,5,6-PENTAHYDROXY-2-HEXANONE): A simple sugar with the molecular formula $C_6H_{12}O_6$ that is found in honey, most sweet fruits and starches that gives fruits their sweet taste. It is the sweetest of all naturally occurring carbohydrates and has low glycaemic value. Fructose, together with glucose and galactose are absorbed directly into the bloodstream during digestion. Crystalline fructose obtained from processing corn or sugar is used in food and beverages as a nutritive.

L'HOPITAL'S RULE (L'HOSPITAL'S RULE): A rule named after the French mathematician, Guillaume de L'Hôpital (1661-1704). It is used in finding the limit of the ratio of two functions of the same variable x, as x approaches a value a. This is equal to the limit of the ratios of their derivatives with respect to x. For example, it can be used to find the limits of $\frac{f(x)}{h(x)}$ at a point in which both $f(x)$ and $h(x)$ are zero and the ratio is therefore indeterminate. This can be stated as $\lim_{x\to a}\frac{f\prime(x)}{h\prime(x)} = L$. For example, to evaluate $\lim_{x\to 0}\frac{\sin 3x}{\sin 2x}$, let $f(x)=\sin 3x$ and $h(x)$ = sin 2x; $f(0)$ = 0 and $h(0)$ = 0, this implies that the ratio is indeterminate, therefore L'hopital's rule is applied by finding the derivatives of $f(x)$ and h(x), which gives $f'(0)$ = 3cos3(0), expressed as $f'(0)$ = 3 and $h'(0)$ = 2cos2(0), which gives $h'(0)$ = 2. Therefore, the $\lim_{x\to 0}\frac{\sin 3x}{\sin 2x} = \frac{3}{2}$.

LA NIÑA: The cooler phase of El Niño, which occurs at irregular intervals in the equatorial Pacific (western coast of South America) regions. It is characterised by cooler surface ocean temperatures, bringing widespread changes in weather patterns across the globe, which can lead to intense storms in some places and droughts in others; but generally its effects are not as extensive and damaging as those of El Niño.

LAB/LABORATORY COAT: Protective clothing, worn to protect clothing when working in a laboratory. It is usually of white colour and loose fitting, extending to or beyond the knee.

LABEL: 1. In chemistry, it is a radioactive element used in a compound to trace the mechanism of a chemical reaction. Alternatively, to make an atom or atoms in a compound radioactive, to use in determining the mechanism of a reaction. 2. In mathematics, to associate some entity with each node of a tree in order to distinguish it from all others. For example, a rooted binary tree can be labelled with binary numbers of which the digits represent the direction taken at each successive node on the route from the root to the associated vertex. Also, the horizontal label across the bottom and the vertical label along the side of a graph indicate the facts listed in it. 3. In computer science, to compute a group of characters, such as a number or a word, appended to a particular statement in a program to allow its unique identification.

LABELLED COMPOUND (RADIO-LABELLED COMPOUND): A compound containing either a radioactive or stable isotope. They are radiochemicals prepared from primary radioisotopes by employing suitable synthetic reactions or labelling procedures. They are used as radiotracers in chemistry, biology, agriculture and medicine.

LABELLED TREE: A tree in which some unique entity labels each node or vertex. Its vertices are usually labelled 1, 2, ..., n.

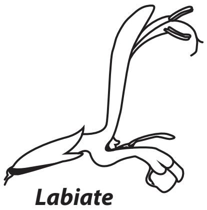
LABELLUM: 1. Also called **lip**, it is the lower of the three petals of an orchid flower, which differs in morphology and patterning from the two lateral petals and gives the flower its characteristic form. The laterals and the sepals may all be similar but the labellum is always distinct and usually much larger than the other perianth segments. It serves as a platform for pollinating insects, which are attracted by its distinctive colours and markings. 2. The platform formed by the lower petal or group of fused petals in various other lipped flowers, such as those of the Labiatae, Scrophulariaceae and Leguminosae.

LABIA MAJORA: The two outer folds of skin that surround the clitoris, the opening of the urethra and the opening of the vagina of human females. Each labium majorum has two surfaces, an outer, pigmented surface covered by pubic hair; and an inner smooth surface containing adipose tissues.

LABIA MINORA (NYMPHAE): Thin membranous hairless folds of skin outside the vaginal opening situated between the labia majora. It extends from the clitoris obliquely downward, forming the hood of the clitoris.

LABIAL: 1. Of, relating to, or situated near the lips or labia. 2. Of, or relating to the folds on the outside of the female sex organs (labia majora or labia minora).

LABIATAE (LAMIACEAE): A large dicotyledonous family, commonly called the mint family, comprising some 3000 species in about 200 genera. Most species are shrubby or herbaceous. Labiates characteristically have stems that are square in cross section and simple leaves in opposite decussate pairs. They are often covered in aromatic hairs and many species, such as mints (Mentha), sage (*Salvia officinalis*) and thymes (Thymus), are used as pot herbs.



Labiate

LABILE: Describing a chemical compound in which certain atoms or groups can easily be replaced by other atoms or groups. The term is applied to coordination complexes in which ligands can easily be replaced by other ligands in an equilibrium reaction.

LABIUM: 1. A lip-like structure of some anthropods, which is the lower lip, formed from a fused pair of appendages and used for biting and chewing. Examples are found in cockroaches and grasshoppers. 2. The lower lip of the corolla of a labiate flower. 3. Any of the two pairs of fleshy fold that surrounds a woman's genital organs, forming part of the vulva.

LABORATORY: A room or building equipped for scientific experiments, research and teaching.